

Product Category Management

Swiggy Instamart E- commerce Database Management System

1.Database Design

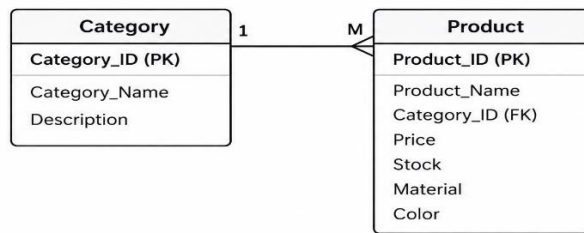
1.1 Category Table

Attribute	Data Type	Constraint
Category_ID	NUMBER	Primary Key
Category_Name	VARCHAR2(50)	NOT NULL, UNIQUE
Description	VARCHAR2(200)	—

1.2 Product Table

Attribute	Data Type	Constraint
Product_ID	NUMBER	Primary Key
Category_ID	NUMBER	Foreign Key
Product_Name	VARCHAR2(100)	NOT NULL
Brand	VARCHAR2(50)	—
Price	NUMBER(10,2)	—
Stock	NUMBER	—
Weight	VARCHAR2(20)	—
Expiry_Date	DATE	—
Image_URL	VARCHAR2(255)	—

2. Primary and Foreign Key Relationship



3 Table Creation

3.1 Create Category Table

```
CREATE TABLE Category (  
2     Category_ID NUMBER PRIMARY KEY,  
3     Category_Name VARCHAR2(50) UNIQUE NOT NULL,  
4     Description VARCHAR2(200)  
5 );
```

Table created.

3.2 Create Product Table

```
CREATE TABLE Product (  
  2     Product_ID NUMBER PRIMARY KEY,  
  3     Category_ID NUMBER,  
  4     Product_Name VARCHAR2(100) NOT NULL,  
  5     Brand VARCHAR2(50),  
  6     Price NUMBER(10,2) NOT NULL,  
  7     Stock NUMBER,  
  8     Weight VARCHAR2(20),  
  9     Expiry_Date DATE,  
 10     Image_URL VARCHAR2(200),  
 11     FOREIGN KEY (Category_ID)  
 12     REFERENCES Category(Category_ID)  
 13 );
```

Table created.

4.Data Insertion

4.1 Category Records

```
SQL> INSERT INTO Category VALUES (101, 'Fruits', 'Fresh fruits and seasonal fruits');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Category VALUES (102, 'Vegetables', 'Fresh vegetables');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Category VALUES (103, 'Beverages', 'Juices and soft drinks');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Category VALUES (104, 'Snacks', 'Chips, biscuits and snacks');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Category VALUES (105, 'Dairy', 'Milk and dairy products');  
  
1 row created.
```

4.2 Product Records

```

SQL> INSERT INTO Product
  2 VALUES (1, 101, 'Apple', 'Fresh Farm', 120.00, 50, '1 kg',
  3          TO_DATE('20-09-2026','DD-MM-YYYY'), 'apple.jpg');

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (2, 101, 'Banana', 'Farm Fresh', 60.00, 80, '1 kg',
  3          TO_DATE('18-09-2026','DD-MM-YYYY'), 'banana.jpg');

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (3, 102, 'Tomato', 'Organic Farm', 45.00, 100, '1 kg',
  3          TO_DATE('16-09-2026','DD-MM-YYYY'), 'tomato.jpg');

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (4, 102, 'Potato', 'Fresh Farm', 50.00, 75, '1 kg',
  3          TO_DATE('25-09-2026','DD-MM-YYYY'), 'potato.jpg');

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (5, 103, 'Orange Juice', 'Real', 110.00, 40, '1 L',
  3          TO_DATE('30-10-2026','DD-MM-YYYY'), 'orange_juice.jpg');

1 row created.

SQL>
SQL> INSERT INTO Product
  2 VALUES (6, 103, 'Coca Cola', 'Coca Cola', 45.00, 60, '750 ml',

```

5.Data Retrieval

5.1 Display Category Records

```
SQL> SELECT * FROM Category;
```

```
CATEGORY_ID CATEGORY_NAME
```

```
-----
DESCRIPTION
-----
```

```

| | 101 Fruits
Fresh fruits and seasonal fruits

```

```

| | 102 Vegetables
Fresh vegetables

```

```

| | 103 Beverages
Juices and soft drinks

```

```
CATEGORY_ID CATEGORY_NAME
```

```
-----
DESCRIPTION
-----
```

```

| | 104 Snacks
Chips, biscuits and snacks

```

```

| | 105 Dairy
Milk and dairy products

```

```
SQL> COMMIT;
```

```
Commit complete.
```

5.2 Display Product Records

```
SQL> SELECT * FROM Product;
```

PRODUCT_ID	CATEGORY_ID	PRODUCT_NAME	BRAND	PRICE	STOCK	WEIGHT	EXPIRY_DA	IMAGE_URL
1	101	Apple	Fresh Farm	120	50			

```
PRODUCT_ID CATEGORY_ID
-----
PRODUCT_NAME
-----
BRAND
-----
PRICE
-----
STOCK
-----
WEIGHT
-----
EXPIRY_DA
-----
IMAGE_URL
-----
1 kg
apple.jpg
20-SEP-26
```

PRODUCT_ID	CATEGORY_ID	PRODUCT_NAME	BRAND	PRICE	STOCK	WEIGHT	EXPIRY_DA	IMAGE_URL
1	101	Apple	Fresh Farm	120	50			

```
PRODUCT_ID CATEGORY_ID
-----
PRODUCT_NAME
-----
BRAND
-----
PRICE
-----
STOCK
-----
WEIGHT
-----
EXPIRY_DA
-----
IMAGE_URL
-----
1 kg
apple.jpg
20-SEP-26
```

6.Product Update

```
SQL> UPDATE Product
  2 SET Price = 130.00
  3 WHERE Product_ID = 1;

1 row updated.
```

```
SQL> UPDATE Product
  2 SET Stock = 150
  3 WHERE Product_ID = 3;

1 row updated.
```

7.Product Deletion

```
SQL> DELETE FROM Product
| 2 WHERE Product_ID = 9;

1 row deleted.

SQL> DELETE FROM Product
| 2 WHERE Product_ID = 10;

1 row deleted.
```

8.Category-wise Product Reports

```
SQL> SELECT
| 2 c.Category_ID,
| 3 c.Category_Name,
| 4 p.Product_ID,
| 5 p.Product_Name,
| 6 p.Brand,
| 7 p.Price,
| 8 p.Stock
| 9 FROM Category c
|10 JOIN Product p
|11 ON c.Category_ID = p.Category_ID
|12 ORDER BY c.Category_ID;
```

CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID
101	Fruits	1
	Apple	
	Fresh Farm	130
		50
101	Fruits	2
	Banana	
	Farm Fresh	60
		80
102	Vegetables	3
	Tomato	

9. Category-Wise Total Product Count

```
SQL> SELECT
  2     Category.Category_Name,
  3     COUNT(Product.Product_ID) AS Total_Products
  4 FROM Category
  5 LEFT JOIN Product
  6 ON Category.Category_ID = Product.Category_ID
  7 GROUP BY Category.Category_Name
  8 ORDER BY Category.Category_Name;
```

CATEGORY_NAME	TOTAL_PRODUCTS
Beverages	1
Dairy	1
Fruits	1
Snacks	2
Vegetables	0